



A Method for Failure Analysis for the 994K Pre-Lubrication System

**Maintenance and
Repair**

Application

**Component Life
Management**

**Component
Renewal (CRC)**

**MARC
Management**

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February 2023
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1.0 Introduction

In January 2020, the new fleet of 994K loaders started operating at the S11D mine, northern Brazil, where the extraction modal is heavily based on the mobile crushers feeding system, in a system called Loading and Carriage.

The 04 994K loaders operating at the S11D mine started their operation as part of a customer effort to increase iron ore production, and increase the flexibility of operations, which at that time were mainly composed of TRUCKLESS.

Since the beginning of the operation, GAPS in the application of the assets were identified, especially in the turbochargers and other Engine components that suffer from the lack of adequate lubrication during starts. The pre-lubrication system automatically enters before starts when one of the following conditions is not present:

- Engine oil pressure more than or equal to **3 kPa. (4 psi)**
- Engine run for the last (02) two minutes.
- The Coolant temperature is more than **50° C (122° F)**

The improper use of the pre-lubrication system can cause catastrophic failures in the turbocharger and accelerated wear in the engine's moving components.

As part of an initiative by the SOTREQ Engineering and Product Support team, to reduce the impacts caused by an operation of this nature, some training plans were implemented to identify possible problems in the system:

- Training Plans.
- Operational Monitoring.
- Negotiations with the Maintenance Team.

We identified a failure in the electric engine, Circuit Breakers, and leaks in the pumps. Tests are currently available for the pump, relay, and electric engine power supply, however, there are no actual tests for the engine or information available about them.

The new approach was made on the Dealer, focusing, on the **Project, People Development, and Process.**

- **PROJECT:** To efficiently control the service life and carry out tests of the components of the pre-lubrication system.
- **PEOPLE DEVELOPMENT:** Identification of operators, with a higher incidence of pre-lubrication system bypass, relating the records to the Operator's Time/Registration, and weekly handling of alarms and analysis of **VIMS** memories with the person responsible for the operation.
- **PROCESS:** Identification of system failures and operation problems, and elaboration of a procedure that seeks the moment to replace system components before failures occur so that potential risks can be prevented.

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It was possible to reduce, and sometimes eliminate **100%** of corrective events in the pre-lubrication system, considerably reducing the impacts of failures related to the pre-lubrication system on the **994K** KPIs.

2.0 Best Practice Description

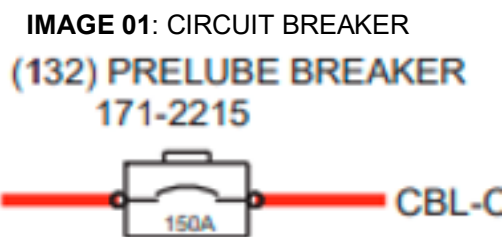
Some equipment fleet 994K Wheel Loaders in S11D mine had several failures in the pre-lubrication system. These failures can be confused with normal situations, in which the system is not activated. This situation is common. An electric engine failure will not generate failure codes.

Early premature failures in the system's engine have made it difficult to solve and detect problems in the lubrication system, as there are no parameters to define the failure. No failure codes show the engine has a problem.

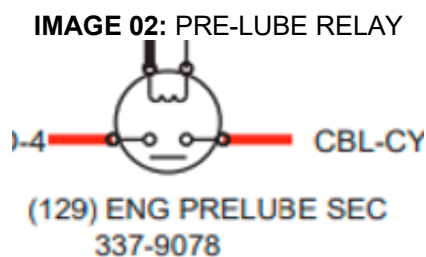
Failures in electric engines are not always simple to detect and solve, the coils do not have resistance close to 0 Ω or overload (indicating open circuit) or close to 0 Ω (0.1 Ω indicating short circuit). There is no parameter for us to effectively define that the electric engine has a problem.

The pre-lubrication system has a standard structure, with the components:

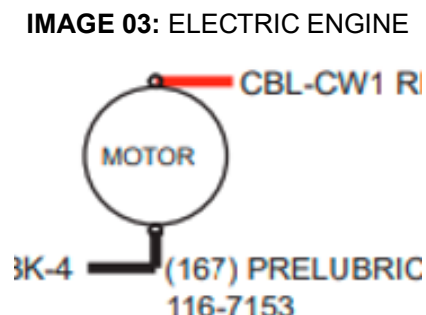
CIRCUIT BREAKER:



ENGINE RELAY:



ELECTRIC ENGINE:



THE PRE-LUBRICATION SYSTEM HAS A STANDARD STRUCTURE, WITH THE COMPONENTS:

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IMAGE 04: SYSTEM SCHEMATIC.

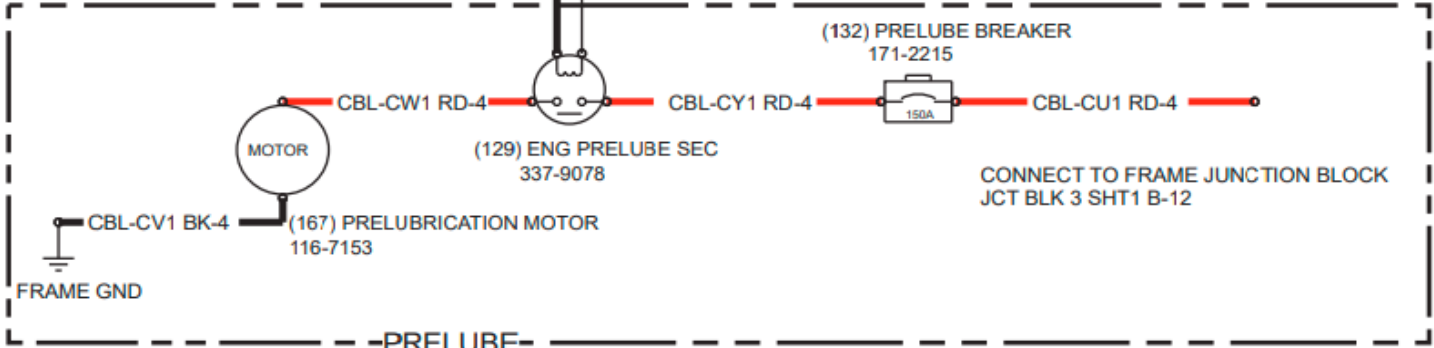
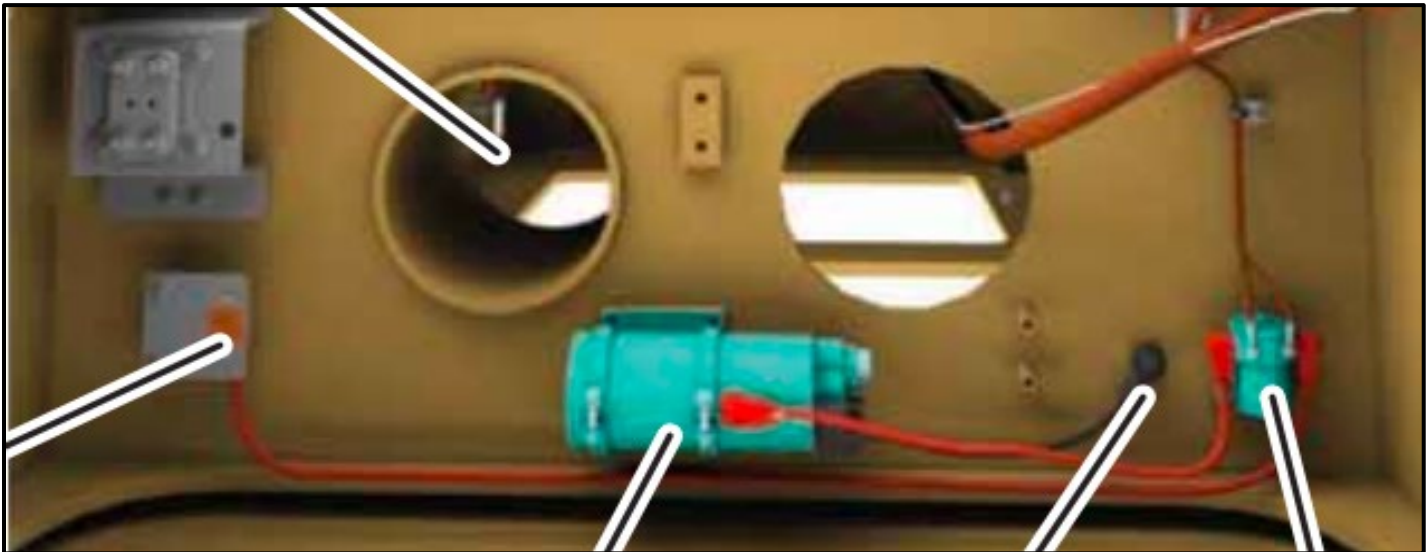


IMAGE 05: CIRCUIT APPLIED TO 994K



Analyzing the structure of a new machine it is possible to define the default parameters to be followed, as the Failure Analysis consists of defining default values and verifying their evolution over time.

The resistance of a cable and a coil can change quickly as they are used, however, the previous analysis of this information can serve as parameters to be compared.

After analyzing the resistance of the defined points, the same points must be checked on the equipment every 250 hours, based on the reference values, the changes in each point are visible.

This data is used to define the best time to replace system components before failure occurs and to be able to quantify the useful life of each component.

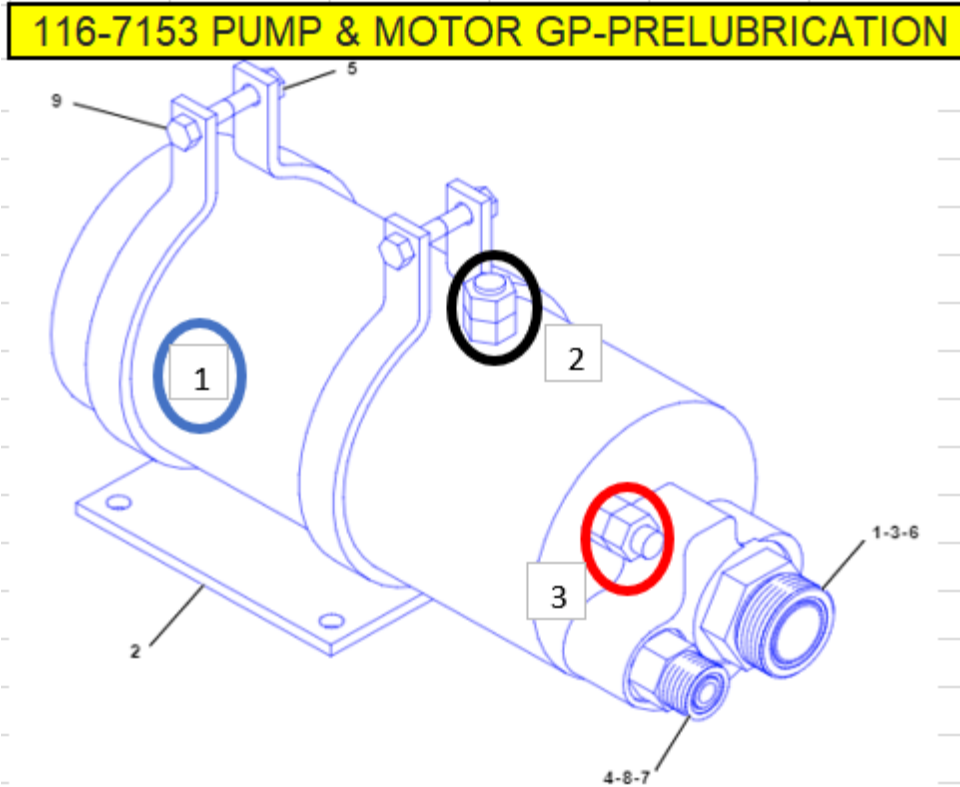
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3.0 Implementation Steps

The first part of the work is evaluating the operational and maintenance conditions of machines and evaluating the failures in the system components and perform the following measurements:

- ✓ RESISTANCE IN ELECTRIC ENGINE.

Image 06: Electric Engine and measuring point



Values for a New Engine:

DESCRIPTION	PART-NUMBER	FIXING POINTS	STANDART VALUE
ENGINE	116-7153	1 - 2	2,6MΩ
ENGINE	116-7153	1 - 3	2,6MΩ
ENGINE	116-7153	2 - 3	0,2Ω

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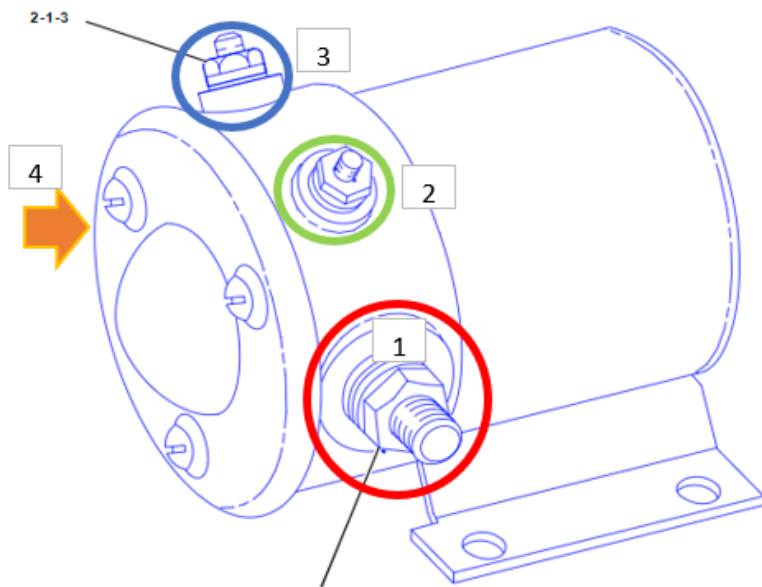
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✓ RELAY RESISTANCE

Image 07: Electrical Relay and Measuring Points
337-9078 SWITCH GP-MAGNETIC

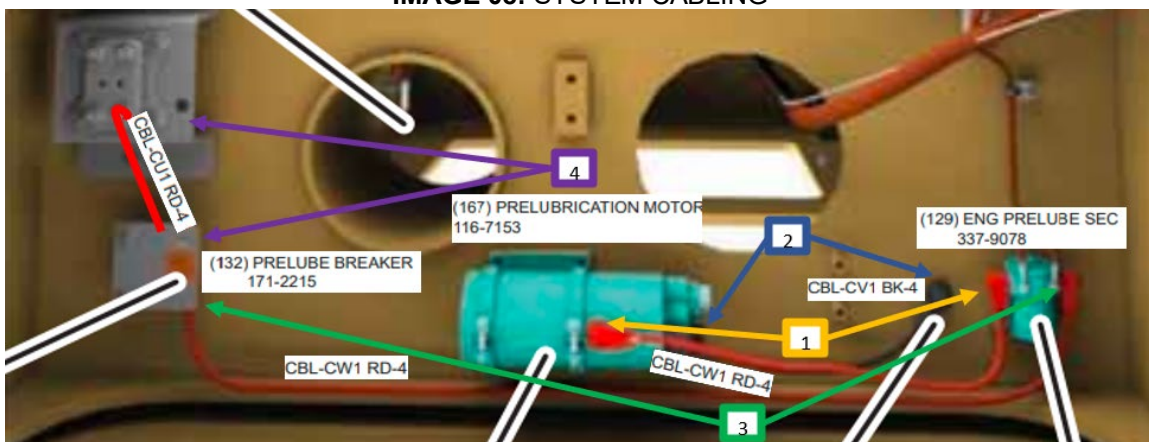


Values for New Relay:

DESCRIPTION	PART NUMBER	FIXING POINTS	STANDART VALUE
RELAY	337-9078	Bobina 2-3	16 Ω +1,6
RELAY	337-9078	Contatos 1-4	0,2 Ω

✓ SYSTEM CONNECTION (CABLES)

IMAGE 08: SYSTEM CABLING



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Values for New Cabling

Relay Cable for Engine (CBL-CW1 RD-4)	472-4095	BETWEEN TERMINALS	0,2 Ω
Motor cable to Ground 2 (CBL-CV1 BK-4)	472-4094	BETWEEN TERMINALS	0,2 Ω
CABLE FROM CIRCUIT BREAKER TO RELAY 3 (CBL-CY1 RD-4)	472-4096	BETWEEN TERMINALS	0,2 Ω
BUS CABLE TO CIRCUIT BREAKER 4 (CBL-CU1 RD-4)	432-4304	BETWEEN TERMINALS	0,2 Ω

After carrying out the measurements and setting the default values, the resistance of the points must be checked every 250 hours, the measurement procedure takes approximately 5 minutes.

All measurements must be made with the equipment off, the parking brake applied, the main switch open and the circuit breaker output cable removed.

It is possible to define the circuit activity and the time to replace the components of the system. In the example, it is possible to verify that the resistance of the electric motors of the 994K (PC30 and PC32) is very low as specified, not enough to be considered short circuits, but low enough to define that the motors are not working and must be replaced.

SYSTEM COMPONENTS				PC30	PC31	PC32	PC33
DESCRIPTION	PART NUMBER	MEASURING POINTS	STANDARD VALUE	VALUE	VALUE	VALUE	VALUE
ENGINE	116-7153	1 - 2	2,6M Ω	55.9Ω	1,222MΩ	95Ω	2,6MΩ
ENGINE	116-7153	1 - 3	2,6M Ω	55.8Ω	2,9MΩ	95Ω	2,6MΩ
ENGINE	116-7153	2 - 3	0,2Ω	0,1Ω	0,2Ω	0,2Ω	0,2Ω
RELAY	337-9078	COIL 2-3	16 Ω+ 1,6	16,1Ω	16,3Ω	16,25Ω	16,1Ω
RELAY	337-9078	CONTACT POINTS 1-4	0,2 Ω	0,1Ω	0,2Ω	0,2Ω	0,2Ω
RELAY CABLE FOR ENGINE 1 (CBL-CW1 RD-4)	472-4095	BETWEEN TERMINALS	0,2 Ω	0,1Ω	0,2Ω	0,2Ω	0,2Ω
CABLE ENGINE TO GROUNDING 2 (CBL-CV1 BK-4)	472-4094	BETWEEN TERMINALS	0,2 Ω	0,2Ω	0,2Ω	0,2Ω	0,2Ω
CIRCUIT BREAKER CABLE FOR RELAY 3 (CBL-CY1 RD-4)	472-4096	BETWEEN TERMINALS	0,2 Ω	0,2Ω	0,2Ω	0,2Ω	0,2Ω
Cable AS for Circuit Breaker (CBL-CU1 RD-4)	432-4304	BETWEEN TERMINALS	0,2 Ω	0,2Ω	0,2Ω	0,2Ω	0,2Ω

IMAGE 09: Resistance Test between points 1-2 of the Electric Engine.

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On the 994K (PC31) motor has the resistance between points 01-02 below normal, the image below shows the moment of measurement. The motor was still working, he could move the pump, however, its useful life was ending, because its internal insulation was bad.

4.0 Benefits

Some benefits can be listed, after applying measurements to the **994K**:

- ⇒ Feedback from mechanics on simple and effective testing of system components.
- ⇒ Removal of components with failures not identified by the machine system and scheduling the replacement of components that are out of the standard.
- ⇒ Assurance that the pre-lubrication system is working.
- ⇒ Accurate lifetime measurement and replacement planning for damaged components.

5.0 Resources Required

The measurement requires a **PN 257-9140 Multimeter**

6.0 Supporting Attachments / References

N/A.

7.0 Related Best Practices

N/A.

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8.0 Acknowledgements

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